

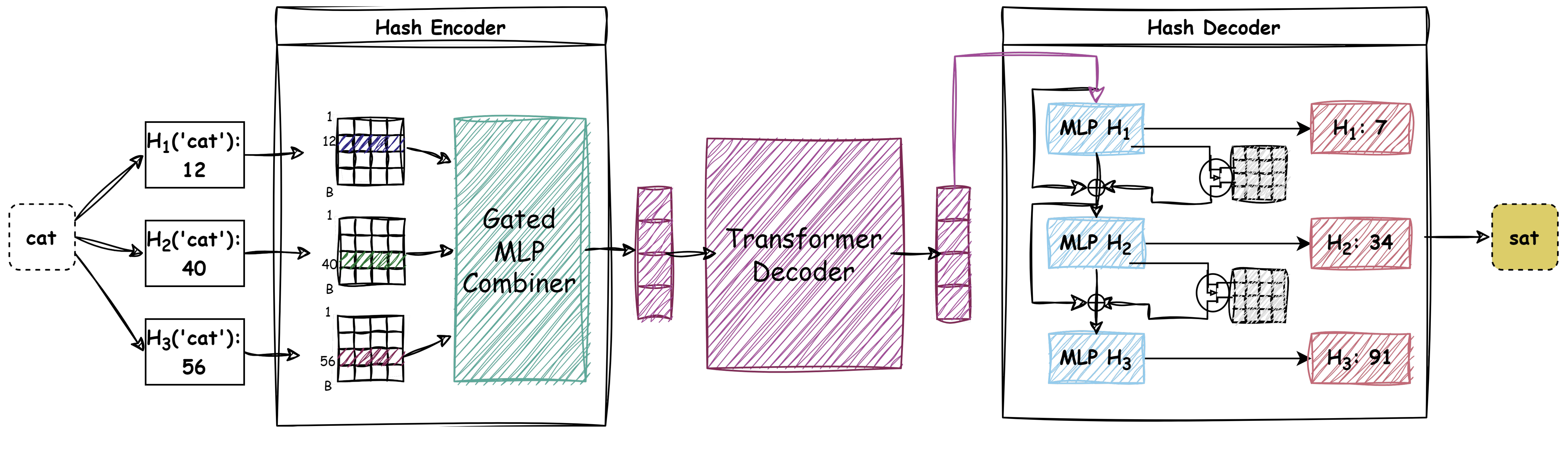
MultiHashFormer: Hash-based Generative Language Models

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ARCHITECTURE · COLLISION-FREE GENERATIVE HASHING



WORKING MECHANISM

HASH ENCODER

• Multi-Hash Indexing

multi-ID signatures for CLM

$$\mathcal{H}_i(w) = \text{MMH3}(w, \text{seed}_i) \pmod{B-1} + 1 \quad \text{if } w \neq \text{pad}.$$

• Gated Compositional Embedding

$$\mathbf{e} = \left(\sum_{i=1}^H \tilde{\alpha}_i \mathbf{E}_{[\mathcal{H}_i(w), :]}^{(i)} \right) \mathbf{W}_s,$$

$$\tilde{\alpha}_i = \frac{\exp^{\alpha_i}}{\sum_{j=1}^H \exp^{\alpha_j}}$$

$$\alpha_i = \sigma \left(\mathbf{E}_{[\mathcal{H}_i(w), :]}^{(i)} \mathbf{W}_1 \right) \mathbf{W}_2.$$

HASH DECODER

• Logit Computation

$$\mathbf{o}^{(i)} = \mathbf{W}_o^{(i)\top} \mathbf{c}^{(i)}$$

• Soft Embedding Retrieval

$$\mathbf{p}^{(i)} = \text{Softmax}(\mathbf{o}^{(i)})$$

$$\mathbf{e}^{(i)} = \mathbf{E}^{(i)\top} \mathbf{p}^{(i)}$$

• State Update via Cascade Mixer

tree-structured recursive networks

$$\mathbf{c}^{(i+1)} = \mathbf{c}^{(i)} + \mathbf{W}_{up}^{(i)\top} \sigma(\mathbf{W}_{dn}^{(i)\top} \begin{bmatrix} \mathbf{c}^{(i)} \\ \mathbf{e}^{(i)} \end{bmatrix})$$

PROBABILITY MODELING

• Training Phase

$$P_{train}(w | \cdot) = \prod_{i=1}^H P(\mathcal{H}_i(w) | \cdot)$$

• Inference Phase

explicitly exclude all unassigned signature

$$P_{inf}(w | \cdot) = \frac{\prod_{i=1}^H P(\mathcal{H}_i(w) | \cdot)}{\sum_{w' \in \mathcal{V}_{actl}} \prod_{i=1}^H P(\mathcal{H}_i(w') | \cdot)}$$

• Higher-Rank Logit Approximation

mitigates the softmax bottleneck

$$\mathbf{A} = [a_{i,j}]_{n \times |\mathcal{V}|}, a_{i,j} = \log p^*(w_j | c_i)$$

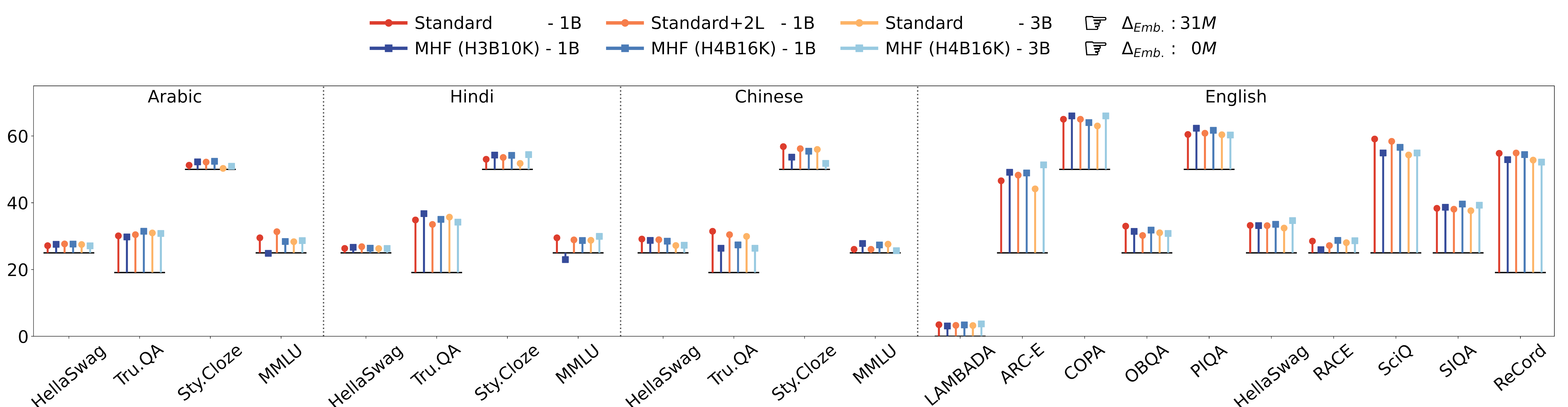
$$r(\hat{\mathbf{A}}_{\text{Stand.}}) \leq d \Rightarrow r(\hat{\mathbf{A}}_{\text{MHF}}) \leq \min(B, Hd)$$

EMPIRICAL RESULTS

SUPERIOR PERFORMANCE OVER STANDARD LMS AT 1B & 3B SCALE

	LAMBADA acc.	ARC-E	COPA	PIQA	HellaSwag	RACE	SIQA	ReCORD	Card-660 ρ
1B	▲4.9	▲4.5	▲7.0	▲0.7	▲2.3	▲2.0	▲2.6	▲1.7	▲0.07
3B	▲8.6	▲6.7	▲3.0	▲2.6	▲3.2	▲1.2	▲2.5	▲1.9	

ZERO-PARAMETER MULTILINGUAL VOCABULARY EXPANSION



MHF supports an upper bound of $B^H = 16000^4$ (65 quadrillion) unique tokens with a constant parameter footprint.

